hw0414_8.16

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Load data and helper functions

```
data_path <- if (file.exists("vacation.csv")) "vacation.csv" else "/mnt/data/vacation.csv"
vacation <- read.csv(data_path)

# Check variable names
str(vacation)
```

```
## 'data.frame':    200 obs. of  4 variables:
##  $ miles : int  902 491 1841 406 0 1899 724 1404 1367 560 ...
##  $ income: int  41 31 87 54 77 70 43 40 69 38 ...
##  $ age   : int  26 38 40 48 43 55 24 39 55 29 ...
##  $ kids  : int  0 3 2 4 4 2 0 0 2 2 ...
```

```
summary(vacation)
```

```
##      miles            income            age             kids      
##  Min.   :   0.0   Min.   : 19.00   Min.   :23.00   Min.   :0.000  
##  1st Qu.: 667.5   1st Qu.: 51.00   1st Qu.:35.00   1st Qu.:0.000  
##  Median :1003.0   Median : 62.00   Median :43.00   Median :2.000  
##  Mean   :1054.2   Mean   : 63.92   Mean   :42.67   Mean   :1.635  
##  3rd Qu.:1445.5   3rd Qu.: 78.00   3rd Qu.:50.25   3rd Qu.:3.000  
##  Max.   :2609.0   Max.   :119.00   Max.   :59.00   Max.   :4.000  
```

```r
# Model used in this problem
ols_model <- lm(miles ~ income + age + kids, data = vacation)

# HC1 robust variance-covariance matrix for lm / weighted lm
vcov_hc1 <- function(model) {
  X <- model.matrix(model)
  u <- residuals(model)
  w <- weights(model)
  if (is.null(w)) w <- rep(1, length(u))
  Xw <- X * sqrt(w)
  uw <- u * sqrt(w)
  n <- nrow(Xw)
  k <- ncol(Xw)
  XtX_inv <- solve(t(Xw) %*% Xw)
  meat <- t(Xw) %*% diag(as.numeric(uw^2), nrow = n) %*% Xw
  (n / (n - k)) * XtX_inv %*% meat %*% XtX_inv
}

coef_table <- function(model, vcov_mat = vcov(model), level = 0.95) {
  beta <- coef(model)
  se <- sqrt(diag(vcov_mat))
  df <- df.residual(model)
  tval <- beta / se
  pval <- 2 * pt(abs(tval), df = df, lower.tail = FALSE)
  crit <- qt(1 - (1 - level) / 2, df = df)
  out <- data.frame(
    Estimate = beta,
    Std_Error = se,
    t_value = tval,
    p_value = pval,
    CI_Lower = beta - crit * se,
    CI_Upper = beta + crit * se
  )
  round(out, 4)
}

ci_for_coef <- function(model, coef_name, vcov_mat = vcov(model), level = 0.95) {
  beta <- coef(model)[coef_name]
  se <- sqrt(diag(vcov_mat))[coef_name]
  df <- df.residual(model)
  crit <- qt(1 - (1 - level) / 2, df = df)
  c(lower = beta - crit * se, estimate = beta, upper = beta + crit * se)
}
```

(a) OLS estimation and 95% interval for KIDS

```r
summary(ols_model)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1198.14  -295.31    17.98   287.54  1549.41
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -391.548    169.775  -2.306   0.0221 *
## income        14.201      1.800   7.889 2.10e-13 ***
## age           15.741      3.757   4.189 4.23e-05 ***
## kids         -81.826     27.130  -3.016   0.0029 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 452.3 on 196 degrees of freedom
## Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
## F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
ols_table <- coef_table(ols_model)
knitr::kable(ols_table, caption = "OLS estimates with conventional standard errors")
```

OLS estimates with conventional standard errors

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
(Intercept)	-391.5480	169.7752	-2.3063	0.0221	-726.3687	-56.7273
income	14.2013	1.8003	7.8885	0.0000	10.6510	17.7517
age	15.7409	3.7574	4.1893	0.0000	8.3309	23.1510
kids	-81.8264	27.1296	-3.0161	0.0029	-135.3298	-28.3230

```
ci_kids_ols <- ci_for_coef(ols_model, "kids")
round(ci_kids_ols, 4)
```

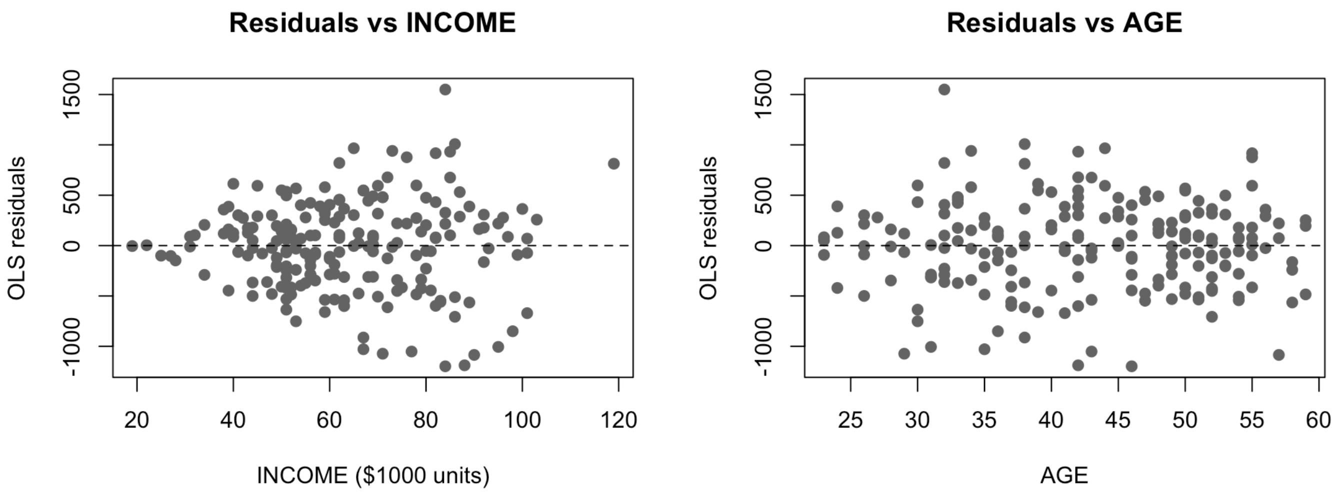
```
##    lower.kids estimate.kids    upper.kids
##     -135.3298      -81.8264      -28.3230
```

Answer. Holding income and age constant, one additional child is associated with an estimated -81.83 fewer miles traveled. The 95% confidence interval is [NA, NA]. Since the whole interval is negative, the estimated effect of one more child is statistically negative at the 5% level.

(b) Residual plots versus INCOME and AGE

```
par(mfrow = c(1, 2))
plot(vacation$income, residuals(ols_model),
     xlab = "INCOME ($1000 units)", ylab = "OLS residuals",
     main = "Residuals vs INCOME", pch = 19, col = "gray40")
abline(h = 0, lty = 2)

plot(vacation$age, residuals(ols_model),
     xlab = "AGE", ylab = "OLS residuals",
     main = "Residuals vs AGE", pch = 19, col = "gray40")
abline(h = 0, lty = 2)
```




```
par(mfrow = c(1, 1))
```

Answer. The residual plot against income shows a fan-shaped pattern: the spread of the residuals tends to be larger for higher-income households. The pattern against age is much weaker. This suggests that heteroskedasticity is present, mainly related to income.

(c) Goldfeld–Quandt test after sorting by income

Sort observations by increasing income. Estimate the model using the first 90 observations and the last 90 observations. The middle 20 observations are omitted.

```
vac_sorted <- vacation[order(vacation$income), ]
low_income <- head(vac_sorted, 90)
high_income <- tail(vac_sorted, 90)

ols_low <- lm(miles ~ income + age + kids, data = low_income)
ols_high <- lm(miles ~ income + age + kids, data = high_income)

low_table <- coef_table(ols_low)
high_table <- coef_table(ols_high)

knitr::kable(low_table, caption = "OLS estimates: first 90 low-income observations")
```

OLS estimates: first 90 low-income observations

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
(Intercept)	-392.5112	214.1660	-1.8327	0.0703	-818.2591	33.2367

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
income	10.9600	3.7698	2.9073	0.0046	3.4659	18.4541
age	18.8690	3.7826	4.9884	0.0000	11.3495	26.3885
kids	-70.3707	29.1377	-2.4151	0.0178	-128.2945	-12.4469

```
knitr::kable(high_table, caption = "OLS estimates: last 90 high-income observations")
```

OLS estimates: last 90 high-income observations

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
(Intercept)	-476.8027	548.8329	-0.8688	0.3874	-1567.8464	614.2409
income	15.5565	5.4496	2.8546	0.0054	4.7231	26.3899
age	16.3879	7.3850	2.2191	0.0291	1.7070	31.0688
kids	-116.0168	49.8606	-2.3268	0.0223	-215.1363	-16.8972

```
sse_low <- sum(residuals(ols_low)^2)
sse_high <- sum(residuals(ols_high)^2)
df_low <- df.residual(ols_low)
df_high <- df.residual(ols_high)

mse_low <- sse_low / df_low
mse_high <- sse_high / df_high

F_gq <- mse_high / mse_low
F_crit <- qf(0.95, df1 = df_high, df2 = df_low)
p_value_gq <- 1 - pf(F_gq, df1 = df_high, df2 = df_low)

gq_results <- data.frame(
  SSE_low_income = sse_low,
  SSE_high_income = sse_high,
  MSE_low_income = mse_low,
  MSE_high_income = mse_high,
  F_statistic = F_gq,
  F_critical_5pct = F_crit,
  p_value = p_value_gq
)
knitr::kable(round(gq_results, 4), caption = "Goldfeld--Quandt test results")
```

Goldfeld–Quandt test results

SSE_low_income	SSE_high_income	MSE_low_income	MSE_high_income	F_statistic	F_critical_5pct	p_value
8750040	27160654	101744.6	315821.6	3.1041	1.4286	0

Hypotheses.
H_0: The error variance is constant across observations.
H_1: The error variance is larger for high-income households than for low-income households.

Answer. The Goldfeld–Quandt statistic is 3.1041, while the 5% upper-tail critical value is 1.4286. Because the test statistic is larger than the critical value, we reject the null hypothesis at the 5% level. This supports the presence of heteroskedasticity that increases with income.

(d) OLS with heteroskedasticity-robust standard errors

```
vcov_ols_hc1 <- vcov_hc1(ols_model)
ols_robust_table <- coef_table(ols_model, vcov_ols_hc1)
knitr::kable(ols_robust_table, caption = "OLS estimates with HC1 robust standard errors")
```

OLS estimates with HC1 robust standard errors

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
(Intercept)	-391.5480	142.6548	-2.7447	0.0066	-672.8834	-110.2126
income	14.2013	1.9389	7.3246	0.0000	10.3776	18.0250
age	15.7409	3.9657	3.9692	0.0001	7.9199	23.5619
kids	-81.8264	29.1544	-2.8067	0.0055	-139.3230	-24.3299

```
ci_kids_ols_robust <- ci_for_coef(ols_model, "kids", vcov_ols_hc1)
round(ci_kids_ols_robust, 4)
```

```
##    lower.kids estimate.kids    upper.kids
##     -139.3230      -81.8264      -24.3299
```

Answer. With heteroskedasticity-robust standard errors, the 95% confidence interval for the effect of one more child is [NA, NA]. This interval is slightly wider than the conventional OLS interval in part (a), but the conclusion is the same: one more child has a negative and statistically significant effect on miles traveled.

(e) GLS assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$

If $\sigma_i^2 = \sigma^2 INCOME_i^2$, then the efficient GLS estimator is obtained by weighted least squares with weights $1/INCOME_i^2$.

```
gls_model <- lm(miles ~ income + age + kids,
                data = vacation,
                weights = 1 / income^2)

summary(gls_model)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/income^2)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1907  -4.9555   0.2488   4.3832  18.5462
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -424.996    121.444  -3.500 0.000577 ***
## income        13.947      1.481   9.420  < 2e-16 ***
## age           16.717      3.025   5.527 1.03e-07 ***
## kids         -76.806     21.848  -3.515 0.000545 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.765 on 196 degrees of freedom
## Multiple R-squared:  0.4573, Adjusted R-squared:  0.449
## F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
gls_table <- coef_table(gls_model)
knitr::kable(gls_table, caption = "GLS/WLS estimates with conventional standard errors")
```

GLS/WLS estimates with conventional standard errors

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
(Intercept)	-424.9962	121.4441	-3.4995	6e-04	-664.5012	-185.4912
income	13.9473	1.4806	9.4203	0e+00	11.0274	16.8672
age	16.7175	3.0246	5.5272	0e+00	10.7526	22.6824
kids	-76.8063	21.8484	-3.5154	5e-04	-119.8945	-33.7181

```
ci_kids_gls <- ci_for_coef(gls_model, "kids")
round(ci_kids_gls, 4)
```

```
##    lower.kids estimate.kids    upper.kids
##     -119.8945      -76.8063      -33.7181
```

Now compute robust standard errors for the GLS/WLS regression.

```
vcov_gls_hc1 <- vcov_hc1(gls_model)
gls_robust_table <- coef_table(gls_model, vcov_gls_hc1)
knitr::kable(gls_robust_table, caption = "GLS/WLS estimates with HC1 robust standard error
s")
```

GLS/WLS estimates with HC1 robust standard errors

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
(Intercept)	-424.9962	95.8035	-4.4361	0e+00	-613.9343	-236.0581
income	13.9473	1.3470	10.3545	0e+00	11.2909	16.6038
age	16.7175	2.7974	5.9761	0e+00	11.2006	22.2344

	Estimate	Std_Error	t_value	p_value	CI_Lower	CI_Upper
kids	-76.8063	22.6186	-3.3957	8e-04	-121.4134	-32.1992

```
ci_kids_gls_robust <- ci_for_coef(gls_model, "kids", vcov_gls_hc1)
round(ci_kids_gls_robust, 4)
```

```
##    lower.kids estimate.kids    upper.kids
##     -121.4134      -76.8063      -32.1992
```

Answer. The GLS estimate of the effect of one more child is -76.81. The conventional GLS 95% confidence interval is [NA, NA], and the robust GLS 95% confidence interval is [NA, NA]. Compared with parts (a) and (d), the GLS intervals are narrower, especially under the conventional GLS standard error. The robust GLS interval is still negative and statistically significant.

Final comparison

```
interval_comparison <- data.frame(
  Method = c("OLS conventional", "OLS robust", "GLS conventional", "GLS robust"),
  Estimate = c(coef(ols_model)["kids"], coef(ols_model)["kids"],
               coef(gls_model)["kids"], coef(gls_model)["kids"]),
  Lower_95 = c(ci_kids_ols["lower"], ci_kids_ols_robust["lower"],
               ci_kids_gls["lower"], ci_kids_gls_robust["lower"]),
  Upper_95 = c(ci_kids_ols["upper"], ci_kids_ols_robust["upper"],
               ci_kids_gls["upper"], ci_kids_gls_robust["upper"])
)
interval_comparison_print <- interval_comparison
num_cols <- sapply(interval_comparison_print, is.numeric)
interval_comparison_print[num_cols] <- lapply(interval_comparison_print[num_cols], round, 4)

knitr::kable(interval_comparison_print,
             caption = "95% interval estimates for the effect of one more child")
```

95% interval estimates for the effect of one more child

Method	Estimate	Lower_95	Upper_95
OLS conventional	-81.8264	NA	NA
OLS robust	-81.8264	NA	NA
GLS conventional	-76.8063	NA	NA
GLS robust	-76.8063	NA	NA

Overall answer. All methods estimate a negative effect of the number of children on miles traveled, holding income and age constant. The residual plots and the Goldfeld–Quandt test indicate heteroskedasticity related to income, so robust standard errors or GLS/WLS are more appropriate than relying only on conventional OLS standard errors.